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Privacy-preserving for user-uploaded images and text in Vision–Language Models

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ABSTRACT

The success of Vision–Language Models (VLMs) in visual question answering and image captioning stems from their ability to align visual data with text instructions effectively. Users often upload image–text pairs to cloud-based VLMs that contain sensitive personal information, making them vulnerable to unauthorized analysis by malicious actors. Traditional data privacy protection approaches, which handle textual and visual modalities independently, are ineffective due to the inherent alignment between image and text. To address this challenge, we propose a user instruction-driven privacy-preserving framework that utilizes a surrogate model to identify and safeguard sensitive images and their associated text. Our experiments on MS-COCO and LLaVA-300K demonstrate that models trained on our privacy-preserving datasets maintain at least 96% of the task performance compared to those trained on unprotected data, while effectively defending against multiple privacy extraction attacks, including reconstruction and attribute inference attacks.

1. Introduction

In recent years, Vision–Language Models (VLMs) have attracted significant attention within the research community. They project visual and textual modalities into a unified vector space, thereby aligning them (Radford et al., 2021). This mechanism enables the model to effectively learn cross-modal information. VLM has demonstrated success across various multimodal tasks, including image classification (Radford et al., 2021), image captioning (J. Li et al., 2023; Girdhar et al., 2023), and image generation (Rombach et al., 2022; Vinker et al., 2022). These services process the user's image–text pairs, leveraging the image content to deliver functionalities such as visual question answering and image captioning in response to the accompanying text instructions. For instance, users may submit brief prompts to receive detailed image descriptions or upload photos from their daily lives to receive guidance and suggestions.

However, this technological advancement brings significant privacy concerns. Users often upload image–text pairs that contain substantial private information, including locations, faces, and more. This private data could be exploited by malicious third parties, leading to privacy breaches. Moreover, the models might inadvertently retain sensitive information in their memory, raising widespread concerns about the unauthorized use of personal data and the potential leakage of user privacy. Therefore, the evolving adversarial threats are crucial for real-world deployment (Deng et al., 2025a).

Currently, numerous privacy protection methods focus predominantly on exclusively safeguarding single-modal data, either text or images, such as faces (Wen et al., 2022), human bodies (Dave et al., 2022; Mauri and Bozzon, 2021), and textual information (Orekondy et al., 2018). The absence of protecting aligned privacy-sensitive content across modalities renders these privacy protections cannot be directly applied to multimodal privacy preservation scenarios. Adversaries can potentially extract sensitive information through unprotected modalities. For instance, an attacker might be able to infer protected content from processed images by analyzing the associated unprotected textual data, thereby compromising the privacy safeguards. Some approaches apply comprehensive protection to image data without considering related textual information, resulting in significant performance degradation in downstream tasks.

Recent attempts to preserve privacy in VLM remain insufficient. Instruction rewriting approaches (Mishra et al., 2025) circumvent image

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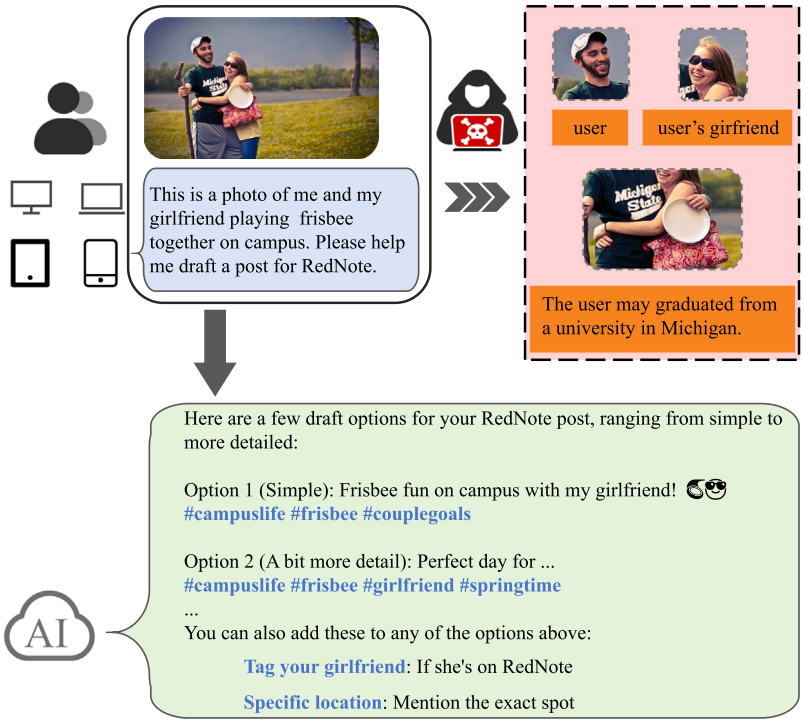


Fig. 1. Users upload image–text data to cloud-based vision–language models, which may expose private information to potential attackers.

transmission by converting visual queries to text commands, fundamentally sacrificing the visual reasoning capacity. Crucially, none of these approaches achieves semantically aligned protection across image and text modalities simultaneously (Sander et al., 2024; Tran et al., 2025).

In this paper, we propose a privacy protection mechanism specifically designed for image–text multimodal data. To align privacy-sensitive information between images and corresponding textual descriptions, we employ a surrogate model that identifies privacy-related elements in both modalities. For image privacy, we implement the standard Block Discrete Cosine Transform (BDCT) methodology employed in JPEG compression (Wallace, 1991). BDCT provides an efficient method for selective visual features while preserving essential image characteristics. To optimize the utility–privacy trade-off for downstream tasks, we implement an image information entropy-based constraint mechanism that regulates the extent of visual feature removal. For text, we refer to the Text-to-Text (T2T) privatization framework (Feyisetan et al., 2020) and enhance it with our semantic neighbor token reinforcement mechanism to maintain semantic coherence.

We evaluate our multimodal privacy-preserving methodology on visual question-answering(VQA) and image captioning tasks. Models trained on privacy-protected datasets achieve remarkable performance, maintaining 96.17% of the original model's effectiveness. To validate the efficacy of our protection mechanism, we conduct comprehensive evaluations using third-party models directly on the protected data. The significant performance degradation indicates the effectiveness of our privacy preservation approach. Furthermore, we systematically evaluate our method's robustness against multiple attacks, including image reconstruction attacks, text embedding inversion attacks, membership inference attacks, and privacy-token prediction attacks. The experimental results demonstrate that our protection mechanism exhibits stronger resistance against all tested attacks than baseline methods, which shows the balanced utility–privacy trade-off of our framework.

Our **contributions** can be summarized as follows:

- We propose a practical privacy-preserving framework for multimodal visual question-answering scenarios that provides personalized privacy protection while maintaining model 96% utility at least.
- By applying BDCT transformation, we preserve useful visual features while removing the most sensitive visual information, maintaining model effectiveness on privacy-protected images. To preserve the privacy of textual information and maintain the semantics of text information, we add neighbor tokens semantic information, which preserves the grammatical structure and contextual coherence in the transformed text.
- Our experimental results demonstrate that protected image–text pairs significantly reduce performance in third-party models and against multiple attacks.

2. Related work

Since (Radford et al., 2021) pioneered the application of contrastive learning to multimodal learning, large models have acquired the capability to process multiple modalities (Kim et al., 2021; Li et al., 2021; Yu et al., 2022; J. Li et al., 2023). In recent years, the performance of multimodal large models on tasks such as captioning and visual question answering (VQA) (Dai et al., 2023; Zhu et al., 2023; Liu et al., 2024a) has impressed the academic community. Today, research is increasingly focused on integrating additional modalities beyond text and images (Girdhar et al., 2023). However, this expansion introduces a range of new privacy challenges that must be addressed.

2.1. Visual data privacy-preserving

Recent work has highlighted various security challenges in large language models (LLM) (Deng et al., 2025a,b), including adversarial attacks, prompt injection, and data poisoning. In visual data privacy protection, numerous methods have been proposed to safeguard privacy by obscuring visual data or adding noise to render the data unlearnable (Aditya et al., 2016; Feng et al., 2019). A notable approach by Li et al. (2024) introduces a dynamic window-based method that determines window sizes based on different image regions' proposed Visual Feature Entropy (VFE), followed by random pixel shuffling within these windows. This adaptive window sizing across different regions balances the privacy and utility of the image.

Additionally, many works employ differential privacy to protect visual features (Dwork, 2006; Abadi et al., 2016). These methods typically blur images by introducing Laplace noise into the image pixels. Wen et al. (2022) applies differential privacy noise to facial features and then reconstructs the facial image using a GAN, maintaining facial recognition accuracy while ensuring robust privacy protection.

The application of Block Discrete Cosine Transform (BDCT) (Wallace, 1991) for image privacy protection has gained significant traction, particularly in facial information protection. A recent research (Xu et al., 2020) proposed converting image data into frequency-domain features for image classification and object detection, experimentally demonstrating the feasibility of neural networks learning frequency-domain features. Building on this foundation, Wang et al. (2022) protects facial privacy while maintaining data utility by removing specific frequency information and reorganizing channels. Duetface (Mi et al., 2022) achieves facial information privacy protection by splitting frequency information into low and high-frequency channels for separate processing on the local and server sides. They eliminate channels based on energy. However, energy only reflects the overall magnitude of pixel values in an image and fails to capture the degree of disorder or complexity within the image. More recently, the study (Jin et al., 2024) has experimentally demonstrated that models can perform facial recognition using just a single frequency channel, which means models can accurately learn distinctive facial features even when an image has lost more than 90% of its frequency information. This method shows remarkable resistance to gradient descent-based face reconstruction attacks.

2.2. Textual data privacy preserving

As large language models become increasingly prevalent, addressing the challenge of preventing privacy leaks through these models is of vital importance. Existing studies concentrate on enhancing privacy during the pretraining (Hoory et al., 2021; Qu et al., 2021; Ding et al., 2024), fine-tuning (Huang et al., 2020; Kerrigan et al., 2020; Yu et al., 2021; Lukas et al., 2023; Shen et al., 2023; Ye et al., 2024), and inference phase (Du et al., 2023; Mai et al., 2023; Zhou et al., 2022; Yuan et al., 2023). Numerous approaches have been proposed to address these challenges. One straightforward method subtitles token before the model input, with subsequent restoration in the output phase (Kan et al., 2023; Chen et al., 2023). Another approach converts privacy-sensitive tokens into feature vectors for direct inference within LLMs' intermediate layers (Qu et al., 2021). The Privacy-Preserving Prompt Tuning (RAPT) (Y. Li et al., 2023) technique employs Text-to-Text (T2T) methods to substitute sensitive tokens while combining prompt tuning and token reconstruction to enhance model utility.

Several studies have investigated the application of homomorphic encryption for preserving text privacy (Chen et al., 2022; Hao et al., 2022; Lu et al., 2023; Zimerman et al., 2023; Liu and Liu, 2023). This approach enables secure computation over encrypted data, theoretically providing robust privacy protection. However, the substantial computational overhead and resource requirements associated with homomorphic encryption operations limit its applicability in real-world scenarios.

2.3. Multimodal data privacy protections

A recent study (X. Liu et al., 2024) pioneered the exploration of multimodal data protection, introducing novel approaches to prevent unauthorized use of personal data on social media. However, this privacy protection approach significantly limits the utility of the data. According to Marouf et al. (2025), user privacy is protected by removing image information from historical dialogues while retaining only the textual content. However, directly deleting the images may influence the model's utility in subsequent conversations. Recent work has also explored privacy preservation through federated learning architectures, such as leakage-resilient aggregation mechanisms for AI-enabled critical infrastructure (Deng et al., 2025c), which protect privacy by keeping data decentralized during model training. Additionally, differential privacy techniques have been applied to LLM fine-tuning (Deng et al., 2025d), providing formal privacy guarantees through noise injection during the training process.

3. Overview

3.1. Threat model

We assume the multi-turn dialogue scenario involving users, cloud-based Vision-Large models (VLMs), and attackers, as illustrated in Fig. 1. In this scenario, users upload images and related textual information to the server through a client application. The server provides accurate and reliable responses to users' instructions. However, the server routinely logs and stores these conversation records, including textual queries and uploaded images.

We assume the third-party attackers may intercept users' image and text data through network interception during transmission or via unauthorized access to server-side storage. They instruct their pre-trained VLMs to analyze image features, like face, human pose, and textual content. Attackers can extract users' private information, such as the relationships between users and people appearing in the images, through VLMs' integrated image and text data analysis.

3.2. Design goals

This paper aims to process image–text data on the user side that safeguards private information so that even if third parties obtain the user's image–text pairs, they cannot effectively acquire the user's private information through pre-trained VLMs. At the same time, the data maintains utility when submitted to authorized cloud-based large language models.

3.3. Approach overview

We present a privacy-preserving approach for cloud-based VLMs that operate on the user side in Fig. 2. In this scenario, both images and text semantics contain privacy information. Protecting privacy-sensitive information across image and text modalities is essential. We first analyze textual content to identify privacy-sensitive information. Subsequently, we employ a surrogate VLM to localize image regions corresponding to the identified textual privacy elements. When users do not provide corresponding textual input, we assume that the model is expected to generate a description of the image content. In our approach, if the user does not upload accompanying text, the local surrogate model generates an image description, which is then used to identify privacy-sensitive regions within the image. Subsequently, the user only needs to upload the privacy-protected image.

By leveraging the surrogate model's alignment capabilities between textual and visual data, we can easily identify regions in images that correspond to privacy-sensitive text. Following this, we implement separate privacy protection measures for textual and visual data. For images, we selectively remove specific frequencies calculated by BDCT (Wallace, 1991) from the image. We employ image information entropy to assess the utility of visual information, guiding the number of frequency components to remove, which effectively balances image utility and privacy. These preserved frequencies are then transformed back to the spatial domain through inverse discrete cosine transform (iBDCT). For text content, we utilize Text-to-Text (T2T) transformation combined with a semantic neighbor token reinforcement mechanism to preserve textual privacy. Finally, the protected image–text pair is transmitted to the cloud-based VLM for standard inference.

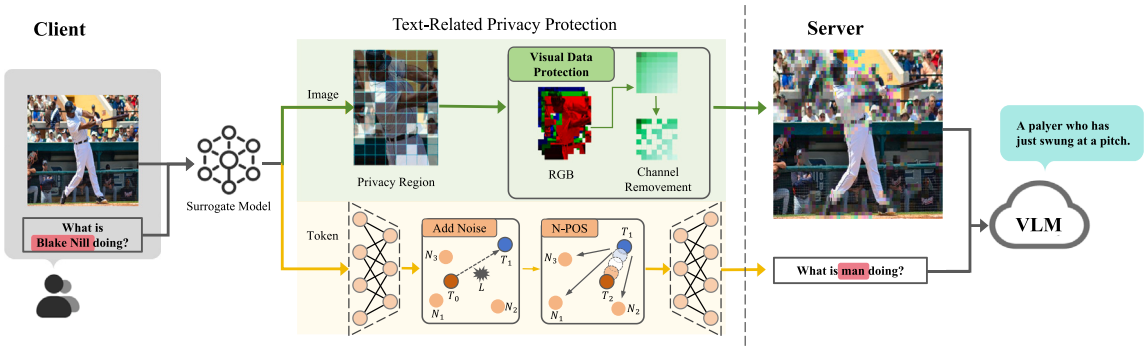


Fig. 2. The framework of our method. First, we analyze the privacy-sensitive tokens in text. Subsequently, we employ a surrogate model to localize the regions in the image corresponding to these tokens. Then we separately protect privacy in the text and image. Finally, the privacy-preserved image–text pairs are uploaded to the server. Token vector T_0 added noise L as T_1, then T_1 is constrained by neighbor vectors N_i. N-POS represents Part-of-Speech Neighbor Constrained.

4. Approach

4.1. Privacy region localization

We leverage the alignment capabilities of a local surrogate VLM to map privacy-sensitive textual entities to their corresponding visual regions via cross-modal attention. This text-guided localization strategy is motivated by three considerations: (1) *User intent alignment*: natural language instructions directly specify privacy targets without manual region annotation; (2) *Efficiency*: text provides compact semantic cues(Radford et al., 2021; S. Liu et al., 2024), avoiding exhaustive image scanning, which would require analyzing all visual content; (3) *Multimodal consistency*: the surrogate VLM's shared embedding space ensures that localization inherently achieves joint text–image protection.

Specifically, we first identify privacy-sensitive tokens t_p in the text by detecting nouns and pronouns using part-of-speech (POS) tagging. This is informed by privacy regulations, including the EU GDPR, ISO/IEC 27701:2019, and China's Personal Information Protection Law (Regulation, 2016; Lachaud, 2020; Yin et al., 2022), which define personal data categories (names, location data, biometric identifiers) that are predominantly represented as nouns and proper nouns in natural language.

For each privacy token t_p, we input the text–image pair into the surrogate VLM, which generates a cross-modal attention score map reflecting the correlation between t_p and various image regions. To mitigate attention-score noise and obtain more comprehensive protection regions, we apply the Gaussian blur (Burt and Adelson, 1987) to smooth the attention maps. We then annotate regions where attention scores exceed a predefined threshold (0.6) and merge the heatmap regions associated with all privacy-related tokens to form the final privacy-sensitive region mask.

4.2. Privacy and utility in visual data

To balance privacy and the utility of visual data, we retain certain visual features in the data by image information entropy. For privacy-sensitive visual data, the information content measured by entropy, which represents the amount of information within a random variable, directly determines what an adversary can extract (Shannon, 1948; Rodriguez-Carrion et al., 2015). Higher entropy corresponds to more information content with clearer contours and vivid colors, and vice versa. Privacy attacks, whether face recognition (Schroff et al., 2015), attribute inference (Fredrikson et al., 2015), or identity linkage, fundamentally rely on discriminative features present in the image. These features manifest as fine-grained details: facial textures (Ahonen et al., 2004), edge sharpness, color variations, and structural patterns encoded primarily in high-frequency components.

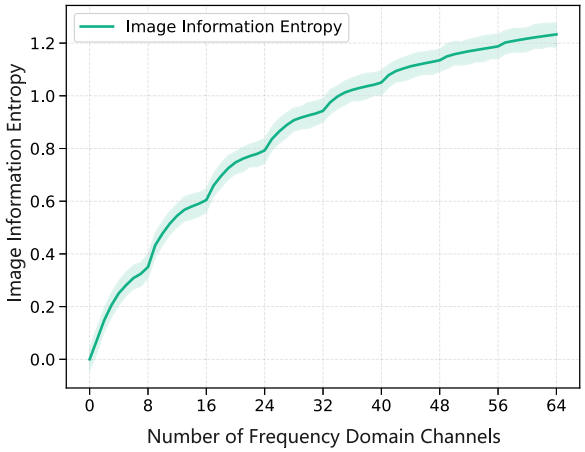


Fig. 3. The trend of the average Image Information Entropy across the MS-COCO dataset.

We depart from traditional approaches (Mi et al., 2022; Ji et al., 2022; Mi et al., 2023) that rely on selecting specific frequency channels or energy-based methods for image privacy protection. Instead, we use image information entropy to standardize image utility while selectively removing different visual features.

After getting the region of the image, it follows the standard BDCT methodology. Initially, each image is preprocessed to a uniform size of 384 × 384 pixels and then divided into 48 × 48 blocks, each of size 8 × 8. Subsequently, we transform each block from the spatial domain to the frequency domain and selectively remove specific channels in the frequency domain.

For a given grayscale image I, the pixel value at position (x, y) is denoted as $I(x, y)$, ranging within the interval $[0, 255]$. We denote $p_{i,j}$ to represent the probability of this pixel value appearing in the image. The information entropy of image I is then defined in (1) as follows:

$$H(I) = -\frac{1}{C}\sum_{i}^{w-1}\sum_{j}^{h-1} p_{i,j}\log p_{i,j}, \tag{1}$$

where C represents the number of image channels, w and h denote the width and height of image.

Images with a greater number of frequency channels exhibit higher information entropy. This relationship can be understood through the mathematical framework of the Discrete Cosine Transform (DCT), which is shown in Eq. (2) (3). As more channels are eliminated, the variation in pixel values diminishes, resulting in lower image entropy. In the extreme case where only the direct current (DC) component remains, the entire block reduces to a single pixel value, causing the

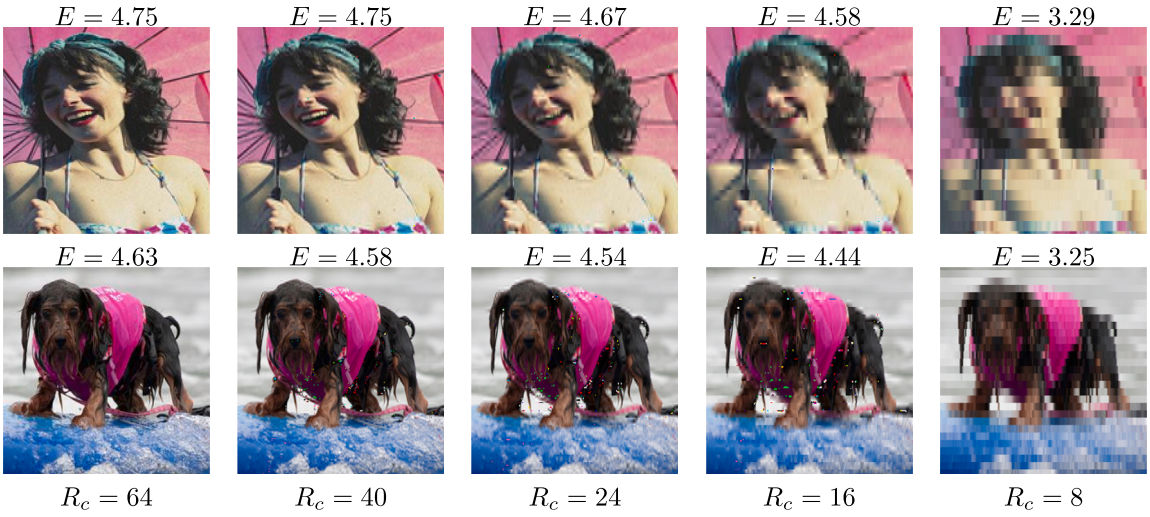


Fig. 4. The image information entropy varies as different frequency channels are removed following the DCT transformation of the image.

image information entropy of that block to reach zero.

$$F_{u,v} = C_u C_v \sum_{x=0}^{N-1} \sum_{y=0}^{M-1} f_{x,y} \cdot G_u(x, N) \cdot G_v(y, M), \tag{2}$$

$$f_{x,y} = \sum_{u=0}^{N-1} \sum_{v=0}^{M-1} C_u C_v F_{u,v} \cdot G_u(x, N) \cdot G_v(y, M), \tag{3}$$

where $f_{x,y}$ denotes the frequency of occurrence of a pixel value at position (x, y) in the image, and $F_{u,v}$ denotes the frequency at position (u, v) in the frequency domain. $G_u(x, N) = \cos(2x+1)u\pi \cdot (2N)^{-1}$ and $G_v(y, M) = \cos(2y+1)v\pi \cdot (2M)^{-1}$ are the basis function of DCT. C_u and C_v are normalization coefficients ensuring orthogonality of the transformation and conservation of energy.

We show the relationship between image information entropy and the number of channels in the frequency domain of the images in the MS-COCO dataset. In Fig. 3, the x-axis represents the number of channels removed, progressing from high to low frequencies. The overall curve indicates that the information content within the images gradually decreases as more channels are removed.

In Fig. 4, we visually demonstrate the relationship between the number of frequency channels R_c and image information entropy E. As the frequency channels are removed, the image information entropy progressively diminishes, resulting in more blurred images.

Before getting frequency channels, we calculate the image information entropy of each image block. These blocks are then transformed into the frequency domain, and frequency channels are systematically removed. The information entropy is recalculated after applying the inverse DCT to restore the spatial domain. The process, shown as Algorithm 1, continues until the entropy value falls below a predetermined threshold.

We perform frequency channel selection independently on the red, green, and blue components for RGB images until their respective information entropy values are reduced below the threshold. Then we shuffle these three components before recombining them, which enhances privacy protection while preserving the structural and texture features.

Algorithm 1 Channel Selection Based on Entropy

Require:
Image I;
Threshold ϵ;
Ensure:
Processed image data I';
$B \leftarrow$ selected_blocks(I)
for b_i in B **do**
  $e_i \leftarrow$ calculate_entropy(b_i)
  $e_i' \leftarrow e_i$
  while $e_i' > e_i \cdot \epsilon$ **do**
    $f_i \leftarrow$ dct_trans(b_i)
    $f_i' \leftarrow$ remove_frequency_channels(f_i)
    $b_i \leftarrow$ idct_trans(f_i')
    $e_i' \leftarrow$ calculate_entropy(b_i)
  end while
end for
$B' \leftarrow$ replace_blocks(b_i, b_i')
$I' \leftarrow$ combine_block(B')
return I'

4.3. Privacy and utility in textual data

Although image data is protected, associated text can still expose private information about the image content to third-party attackers, such as the presence of individuals or the user's environment. To comprehensively safeguard user privacy in both visual and textual data, we add Laplace noise to each privacy token vector from users' queries, which is intended to remove privacy-sensitive textual features

$$\hat{v}_i = v_i + \eta, \tag{4}$$

v_i represents the embedded vector from the token t_i.

However, the noise η may lead to $\hat{v}_i$ deviating from the original semantics, resulting in degraded downstream task performance. Inspired by Y. Li et al. (2023), we implement a constraint mechanism based on neighboring vectors with identical parts of speech (POS) to mitigate this. First, within the vector space $\mathbb{R}^{n \times d}$, we retrieve m nearest neighbors with matching POS tags for vector v_i. These neighbors are represented as $c = [c_1, c_2, \ldots, c_m] \in \mathbb{C}$, where $\mathbb{C}$ denotes the set of tokens sharing the same POS tag as x_i. After obtaining the privatized representation v_i by adding Laplacian noise, we compute a weighted combination with all vectors in c as the final vector v_i'. The noise Mechanism can be expressed as:

$$v_i' = w_0 v_i + \sum_{i=1}^{m} w_i E(c_i), \tag{5}$$

w_i represents predefined weights. The final replacement token x' is obtained through the nearest neighbor search in the vector space.

The proposed constrained noise within the text's vector space ensures text privacy at the token level. And it allows the text, once encoded into a vector by a cloud-deployed VLM, to maintain high semantic fidelity with the original. Subsequently, the fine-tuned VLM remains capable of processing users' textual instructions to provide answers, thus preserving privacy without significantly compromising utility.

5. Experiments

5.1. Models and datasets

VLMs. We evaluate our method across diverse VLMs, including the early model BLIP2 (J. Li et al., 2023) and several state-of-the-art open-source multimodal models: Llama-3.2-Vision-Instruct (Dubey et al., 2024), LLaVA-v1.5-7B (Liu et al., 2024b), and Qwen2-VL (Wang et al., 2024), fine-tuned using the LLaMA-factory framework (Zheng et al., 2024). We employ CLIP, BLIP, and BLIP2 as surrogate models in our method. These surrogate models are used exclusively to localize image regions corresponding to textual content for privacy protection. Importantly, our approach does not impose strict performance requirements on the surrogate models. It is sufficient that their image–text alignment capabilities exceed CLIP.

Datasets. Our experiments use four datasets: MS-COCO (Chen et al., 2015), LLaVA-300K (Liu et al., 2024b), PhotoChat++ (Lee et al., 2024) , and Conceptual-Captions (Sharma et al., 2018).

• *Image Captioning Task.* From the image captioning task, we select 16,000 images from the MS-COCO dataset, each paired with five captions. We allocate 12,000 images and their corresponding captions to the training set, while the remaining 4000 images are split evenly into 2000 for validation and 2000 for testing. The MS-COCO dataset is primarily used for caption-related tasks.

• *Visual Question Answering Task.* For the visual question answering (VQA) task, we employ 6000 images and their questions from the LLaVA-300K, PhotoChat, and Conceptual-Captions.

LLaVA-300K is an academically image–text dataset, making it suitable for both model training and comprehensive quantitative evaluation. We split the dataset into 80% for training and 20% for testing. Each image in the dataset was paired with five rounds of dialogue interactions, ensuring comprehensive coverage of diverse conversational scenarios.

PhotoChat++ is a publicly available multi-modal dialogue dataset, which contains six intent labels, a triggering sentence, an image description, and salient information (e.g., "words" or "phrases"). We select the images and their descriptions in our experiments.

Conceptual Captions (Sharma et al., 2018) is a large-scale web-harvested dataset comprising image–caption pairs, where captions are automatically extracted from HTML and processed through an automated filtering pipeline to ensure quality and diversity.

Medical and biometric imaging datasets are excluded from our evaluation, as they operate under distinct regulatory frameworks and require specialized domain-specific privacy techniques beyond the scope of the general vision–language framework.

Evaluation Metrics. We evaluate five models on the image captioning task and VQA, employing multiple metrics including BLEU-4, ROUGE-1, ROUGE-2, and ROUGE-L to assess the similarity of generated texts and original texts. BLEU-4 (Papineni et al., 2002) measures 4-gram precision (word sequence overlap) between generated and reference captions. ROUGE (Lin, 2004) measures n-gram recall: ROUGE-1 (unigrams), ROUGE-2 (bigrams), ROUGE-L (longest common subsequence).

5.2. Training setup

To make VLMs effectively learn the protected image–text pairs while maintaining fundamental capabilities, we fine-tune the last vision layer and the last LLM layer of the cloud-based VLMs and keep other layers frozen. For the image–text data protection phase, we directly utilized publicly available models.

For our experimental setup, we employ different GPU configurations to meet the requirements of each model. A single NVIDIA GeForce RTX 3090 GPU is used as the testbed for the BLIP-2, Qwen2-VL-2B, and Llama-3.2-11B-Vision-Instruct models. For the LLaVA-v1.5-7B and Qwen-VL-7B models, we utilize six NVIDIA GeForce RTX 4090 GPUs. We configure the learning rate to 1×10^{-5} with the Adam optimizer, complemented by a cosine learning rate and weight decay scheduler. We set the threshold for image information entropy $\epsilon = 0.5$, ensuring the usability and privacy of image data.

5.3. Visual and textual data utility

5.3.1. Image captioning task

Since users may upload only images without text, we assume that they intend for the model to generate descriptions of the images. Therefore, it is necessary to evaluate the model's performance on the image captioning task using privacy-protected images. Since the image captioning task only requires input instructions that describe the image, we use the ground-truth captions from the dataset to localize privacy-sensitive regions in the image. The ground truth in the COCO dataset typically contains fewer than 20 words. However, for models such as Qwen2 and Llama3, directly prompting them to describe the image content often results in excessively long responses, which get very low scores. To address this, we constrain the output to less than 15 words in our prompts.

As shown in Table 1, the models' performance on privacy-protected images is comparable to their performance on raw images. Notably, since we fine-tuned the image encoder component, the models successfully learned the visual features of the privacy-protected regions, thereby ensuring the accuracy of the generated captions.

5.3.2. Vision question answer task

We evaluate our privacy-preserving method on Visual Question Answering (VQA) tasks (Yu et al., 2023), which require sophisticated multimodal understanding and reasoning. We specifically assessed the VQA performance of three state-of-the-art models, such as Qwen2, LLaVA-1.5, and Llama-3.2-vision. We assume that the VQA tasks represent an open-ended scenario, where merely comparing predicted results to the ground truth is insufficient to assess their quality. Therefore, we adopt the evaluation method proposed in Yu et al. (2023) that employs GPT-4. Specifically, we provide predefined prompts, ground truth, and model outputs to GPT-4o, allowing it to score the model's predictions depending on the ground truth.

Our method demonstrates robust performance, as shown in Table 2. Specifically, when models are fine-tuned using data processed by our approach, they achieve, on average across four distinct models, 96.17% of the GPT metric score obtained by counterpart models fine-tuned directly on the raw image–text pairs (baseline, $\eta = \infty$ for text feature noise and raw images). This indicates a negligible impact on efficacy. We also use the metrics in image captioning tasks in Table 1, which comprehensively validate the utility of our approach. These results demonstrate that models successfully learned to extract and utilize meaningful features from privacy-protected images and texts.

To further validate the generalizability of our approach, we evaluate on PhotoChat++, a social media dataset and Conceptual Captions (Sharma et al., 2018). Unlike LLaVA-300K, these datasets contain informal, colloquial language where image captions may not directly describe the visual content. In such settings, token-matching metrics (e.g., BLEU, ROUGE) are unreliable for semantic quality, as they fail

Table 1
Comparative image captioning performance (Our Method vs. Raw Image–Text)

Models	Our method				Raw Image–Text			
	BLEU-4	ROUGE-1	ROUGE-2	ROUGE-L	BLEU-4	ROUGE-1	ROUGE-2	ROUGE-L
BLIP2	35.59	81.65	43.22	50.59	45.60	77.69	43.62	49.16
Qwen-2b	36.57	41.29	12.89	36.90	33.15	32.58	8.51	28.41
Qwen-7b	36.86	41.97	13.43	37.24	32.39	37.26	10.28	33.61
Llava1.5	37.23	41.64	13.07	37.05	36.64	41.31	12.87	36.61
Llama3.2	29.92	36.10	8.96	31.24	29.89	35.67	8.87	30.85

Table 2
VQA performance comparison on LLaVA-300K. **Raw Image–Text**: original unprotected data processed by the target VLM. **Our Method**: privacy-protected data processed by fine-tuned VLMs. **Protected**: privacy-protected data directly processed by untrained off-the-shelf VLMs (adversary attack), simulating an attacker who has no access to the protected data distribution.

Method	Models	BLEU-4	ROUGE-1	ROUGE-2	ROUGE-L	GPT-4 score
Our method	Qwen-2b	50.57	46.87	20.89	32.78	66.89
	Qwen-7b	49.12	49.71	25.82	35.87	62.08
	LLaVA-1.5	43.74	45.48	23.35	31.78	54.51
	Llama3.2-V	31.73	37.73	18.40	30.01	61.26
Raw Image–Text	Qwen-2b	47.07	46.40	22.98	33.52	69.52
	Qwen-7b	53.61	47.43	22.38	32.62	62.93
	LLaVA-1.5	44.32	41.20	23.58	30.66	57.98
	Llama3.2-V	29.40	37.63	14.29	24.39	64.05
Protected	Qwen-2b	24.00	28.59	8.58	18.82	22.48
	Qwen-7b	24.31	29.82	9.57	19.52	30.00
	LLaVA-1.5	29.53	30.03	11.49	20.29	36.93
	Llama3.2-V	23.34	29.68	8.94	19.18	31.88

Table 3
VQA performance on PhotoChat++ and Conceptual Captions evaluated by GPT-4 Score.

Model	Raw Image–Text	Our method	Protected
Qwen-2b	73.45	69.91	22.54
Qwen-7b	78.79	77.34	32.87
LLaVA-1.5	55.36	53.04	35.91
Llama3.2-V	75.44	72.97	33.37

to capture the flexibility of social media text. Therefore, we adopt the GPT-4 score as the evaluation metric for these datasets.

As shown in Table 3, applying privacy protection without fine-tuning leads to a dramatic performance drop across all models. In contrast, fine-tuned models achieve scores of 69.91 and 77.34, which maintains 96.54% utility on average. These show that our method maintains strong utility across diverse data distributions.

5.4. Visual and textual data privacy

5.4.1. Multimodal attack evaluation

To comprehensively assess the robustness of our privacy protection method against realistic threats, we evaluate two types of multimodal attack scenarios where adversaries can leverage both image and text simultaneously.

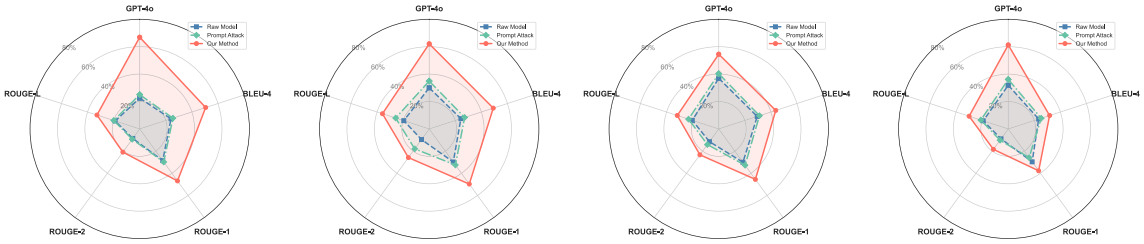
We assume attackers have access to a VLM with comparable performance to the target model and can analyze protected image–text pairs to extract users' privacy. We design two attack strategies: (1) Direct inference attack: using VLMs to directly analyze protected data without modification, and (2) Adversarial prompting attack: employing carefully crafted instructions to guide VLMs in actively searching for and inferring sensitive information. For the adversarial prompting attack, we design prompts that instruct the model to recover hidden information, such as: **"This data has been privacy-protected. Reverse or infer the original sensitive content. Use your reasoning capabilities to recover hidden information from any available clues".** This simulates realistic scenarios where sophisticated adversaries use prompt engineering to bypass privacy protections. We evaluate these attacks on the VQA task, which requires deeper vision–language reasoning compared to image captioning and multimodal retrieval, making it a more stringent benchmark for assessing privacy leakage.

The results in Fig. 5 demonstrate the effectiveness of our method against both attack types. For direct inference attacks, the attacker's VLMs exhibit significantly degraded performance on privacy-protected data compared to original data, indicating that protected image–text pairs are difficult to analyze without additional training. For adversarial prompting attacks, while carefully crafted prompts do lead to modest performance improvements over the baseline, these gains remain minimal and are substantially lower than those achieved by VLMs specifically fine-tuned on privacy-protected data. More importantly, even with adversarial prompting, all models perform significantly worse on privacy-preserving data than on original unprotected data Table 2, demonstrating that our method effectively resists intelligent multimodal attacks and maintains strong privacy protection guarantees.

5.4.2. Attacks for images

In the Threat Model, we consider the attacker attempts to exploit generative models to reconstruct privacy-protected images and extract users' private information. Therefore, we employed state-of-the-art diffusion models to reconstruct the protected images.

Real-ESRGAN (Wang et al., 2021) is a model designed to synthesize high-resolution images from low-resolution inputs, trained on DIV2K (Agustsson and Timofte, 2016) and Flickr2K (Timofte et al., 2018) datasets. The model employs its high-order degradation process and U-Net discriminator to enhance image details while removing annoying artifacts from most real-world images. A recent study (Conde et al., 2025) proposed the first approach, InstructIR, to restore images guided by human-written instructions. This method performs state-of-the-art tasks, including image denoising, deraining, deblurring, dehazing, and low-light image enhancement. We selected these two methods to simulate potential privacy reconstruction



(a) Qwen-2b (b) Qwen-7b (c) Llava1.5 (d) Llama3.2

Fig. 5. Comparative Analysis of Adversarial Inference on Protected Data and Our Method Across Multiple Metrics.

Table 4
Privacy protection against restoration attacks (Real-ESRGAN and InstructIR).

Attack		Real-ESRGAN				InstructIR			
Metric		SSIM ↓	PSNR ↓	MSE ↑	COS ↓	SSIM ↓	PSNR ↓	MSE ↑	COS ↓
Our Method		**0.7170**	**21.7571**	**437.36**	**0.9793**	**.7537**	**22.6360**	**355.72**	**0.9821**
DP	$\eta = 0.5$	0.7239	23.6764	280.87	0.9908	0.8720	28.7792	85.71	0.9978
	$\eta = 5$	0.8865	28.1141	100.95	0.9970	0.9649	38.4548	9.38	0.9996
	$\eta = 10$	0.8900	28.2298	98.35	0.9970	0.9670	39.4178	7.57	0.9996
	$\eta = 50$	0.8900	28.2296	98.35	0.9970	0.9670	39.4077	7.58	0.9996
FaceObfuscator		0.5223	12.9192	3325.27	0.8629	0.5761	13.4607	2934.11	0.8633

Table 5
GPT score and ASR performance across different ϵ Values.

ϵ	GPT score	Attack Success Rate (ASR)							
		Realesrgan				InstructIR			
		SSIM	PSNR	MSE	COS	SSIM	PSNR	MSE	COS
0.3	0.3525	0.4911	15.2753	953.2747	0.8343	0.5345	16.3533	1534.4443	0.8244
0.5	0.6118	0.7170	21.7571	437.3557	0.9793	0.7537	22.6360	355.7179	0.9821
0.7	0.6172	0.7792	23.4262	307.5823	0.9810	0.7974	24.3755	194.7584	0.9855
0.9	0.6295	0.8034	25.3947	102.2811	0.9824	0.8241	26.4895	15.3945	0.9874

attempts by malicious attackers to test the robustness of our privacy protection approach against state-of-the-art reconstruction methods.

The similarity between reconstructed images and their originals was assessed using multiple complementary metrics: Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR) (Johnson, 2006), Structural Similarity (SSIM) (Wang et al., 2004), and Cosine Similarity (COS) (Radford et al., 2021). MSE, PSNR, and SSIM calculate the difference between two images in pixels. COS describes the cosine similarity between the reconstructed image and the original image in the same feature space. Specifically, a higher MSE, a lower PSNR, a lower SSIM, and a lower COS indicate a lower similarity between the reconstructed image and the original image, which means a stronger defense.

Experimental results demonstrate that our method effectively defends against these attacks. As highlighted in Zhao et al. (2024), generative models typically recover high-frequency details in the early stages of restoration. By removing high-frequency and low-frequency information from the images, our method makes it more challenging for generative models to reconstruct the original image.

To validate the correlation between entropy reduction and privacy-preserving, we conduct sensitivity analysis over different entropy thresholds $\epsilon \in \{0.3, 0.5, 0.7, 0.9\}$, where ϵ represents the target entropy as a fraction of the original image entropy. We evaluate each threshold on 500 randomly sampled images from the VQA task, which reflects real-world multimodal application scenarios. For each threshold, we measure: (1) Attack Success Rate (ASR) under reconstruction attacks, (2) VQA performance using GPT-4o.

The results in Table 5 demonstrate that as entropy decreases, privacy preserving strengthens while task utility gradually declines. Notably, the $\epsilon = 0.5$ threshold achieves an optimal balance. The correlation between entropy reduction and both GPT Score and ASR validates that entropy reduction through frequency-domain filtering effectively limits information leakage to adversaries rather than merely degrading visual quality.

5.4.3. Attacks for text

We evaluate the privacy protection capabilities of text protection using three different attack methods. We implement Token Embedding Inversion Attack and Attribute Inference Attack referenced to Y. Li et al. (2023), Mai et al. (2023). Additionally, we employ a black-box attack based on the BERT model to simulate an attacker.

The Token Embedding Inversion Attack uses a nearest-neighbor algorithm to determine the corresponding token for a protected vector. The Attribute Inference Attack means inferring the attributes of the text based on its content. The black-box attack based on BERT involves the attacker predicting the original token from the input text using the BERT model.

We adopt the attack success rate (ASR) as our evaluation metric, which means the attack successfully infers the original token. The experimental results in Fig. 6 indicate that for both the BERT Attack and the Token Embedding Inversion Attack, the attack accuracy drops to nearly 0% when $\eta \leqslant 2$, which corresponds to our chosen parameter settings for text protection. This highlights the robustness of our approach against textual attacks and maintains the utility of our method. For the Attribute Inference Attack, our privacy protection significantly reduces the prediction accuracy, with the accuracy not reaching zero likely, since we only protect nouns in the text to preserve the model's performance.

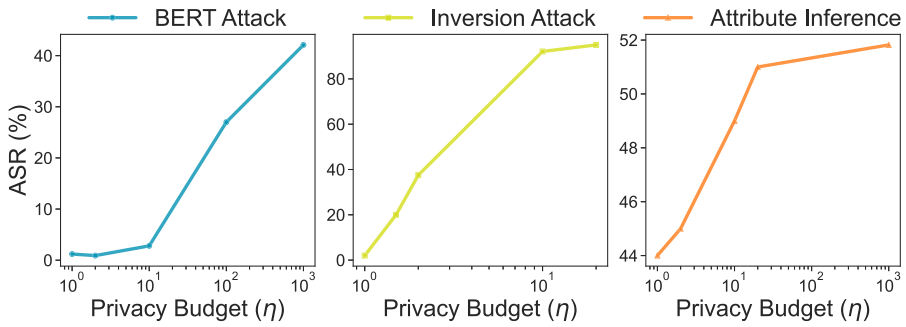


Fig. 6. Accuracy of Three Types of Text Attacks Under Different Values of η.

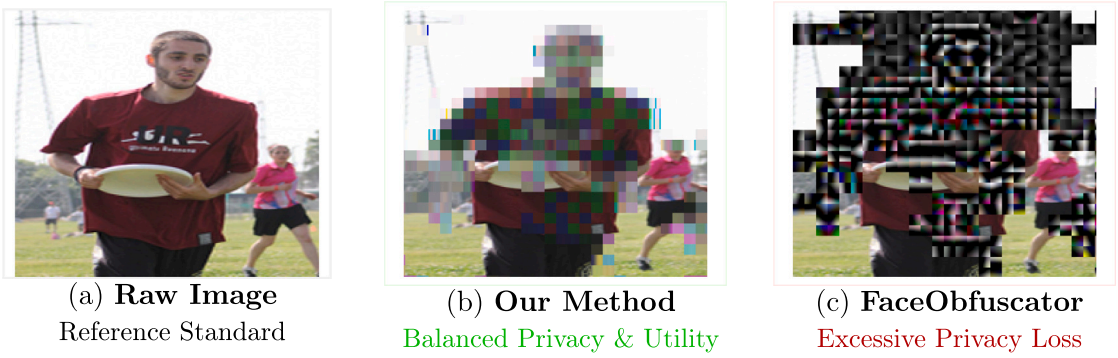


Fig. 7. Visual Comparison of Privacy Protection Methods in VQA Scenarios.
Our method (7(b)) preserves critical visual features for VQA tasks while providing effective privacy protection, compared to the raw image (7(a)) and over-redacted FaceObfuscator (7(c)), which severely degrades utility. **Red arrows** indicate privacy-sensitive regions retained in our approach but obscured by FaceObfuscator.

5.5. Compare with baseline

As there are currently no existing privacy protection methods specifically designed for multimodal question answering, we conduct separate baseline comparisons for visual and textual data. When evaluating image baselines, we apply consistent privacy protection parameters to the text component, and vice versa for text baselines.

5.5.1. Compare with visual baseline

The development of visual privacy protection mechanisms for image captioning and VQA is still in an early stage. We adopt FaceObfuscator (Jin et al., 2024) as our baseline despite its narrow focus on facial obfuscation rather than holistic image protection, as both our method and FaceObfuscator selectively remove frequency-domain features to preserve privacy. However, experimental results demonstrate that our approach achieves significantly higher practical utility. FaceObfuscator's excessive feature removal critically impairs the meaningful visual representations. In contrast, our method has an effective trade-off between privacy and utility. As shown in Fig. 8 and Table 4, it is evident that our method achieves the most balanced privacy–utility trade-off.

For the Differential Privacy (DP) method, the reconstructed images show high similarity to the original images when subjected to image reconstruction attacks. This is because the dp noise mechanism in images is highly similar to the denoising mechanism of diffusion models when generating images. Additionally, the DP method scores lower on GPT-4o evaluations than our method. For FaceObfuscator, this approach performs well in resisting reconstruction attacks due to its removal of excessive visual features Fig. 7. However, precisely because of this, it scores significantly lower than our method, resulting in a substantial loss of data utility shown in Fig. 8.

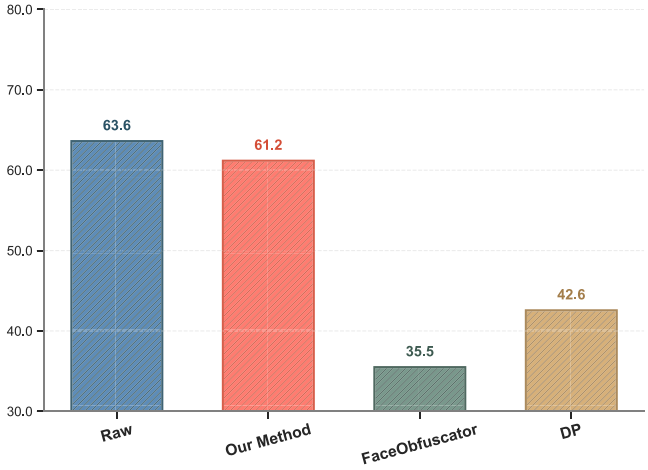


Fig. 8. GPT-4o Score Comparison with Visual Data Privacy-Preserving Baseline Methods.

5.5.2. Compare with textual baseline

We compare our method with RAPT (Y. Li et al., 2023) and traditional differential privacy for text privacy protection methods. Both methods implement privacy protection by adding noise to the vector representations of tokens after embedding. Fig. 9 shows that our method achieves higher scores under equivalent noise levels. When $\eta = 1$, the performance difference among the three methods is relatively small. However, at $\eta = 2$, our method demonstrates a clear advantage,

Table 6
Text privacy protection examples.

Original query	Protected query
What are the ballplayers holding in their fingers?	What are the Bobby_Howry_rhp holding in their equally_culpable?
What is the woman wearing in the image?	What is the newboy wearing in the image?

Privacy tokens (red) are substituted with semantic neighbors (blue).

Table 7
Effect of Surrogate model.

Surrogate model	GPT-4 score↑
CLIP	**61.18**
BLIP	60.93
BLIP2	58.87
w/o	44.25

Table 8
Varying η Impact on model GPT score.

Models	η				
	1	2	10	20	∞
Qwen-2b	36.53	66.89	65.86	69.06	69.52
Qwen-7b	43.07	62.08	62.93	60.04	62.93
Llava1.5	36.73	54.51	54.63	59.92	57.98
Llama3.2-v	34.85	61.26	61.84	64.05	64.05

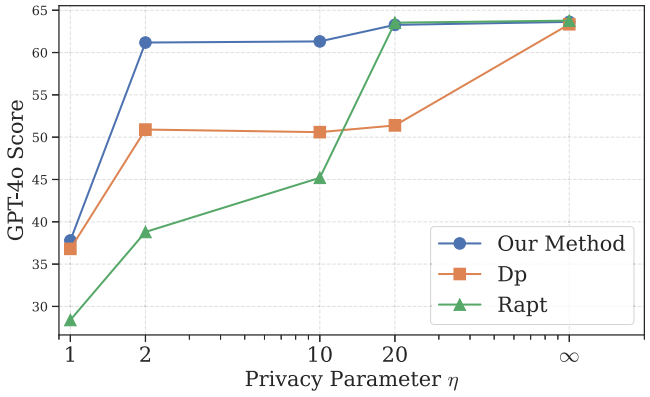


Fig. 9. Performance Comparison with Textual Data Privacy-Preserving Baseline Methods.

achieving the highest score. This result confirms that incorporating neighboring token semantics effectively mitigates the semantic degradation caused by higher noise levels. Accordingly, our method adopts $\eta = 2$ as the parameter setting for text protection.

5.5.3. Qualitative analysis

Fig. 10 presents examples showing how our framework identifies and protects privacy regions. The attention heatmaps confirm that the surrogate VLM's cross-modal attention consistently localizes semantically meaningful sensitive content as specified by the user's instruction, and the corresponding protected images demonstrate effective obfuscation while preserving non-sensitive visual features. Table 6 further illustrates text protection, where privacy-sensitive tokens are substituted with semantic neighbors to prevent token-level identity inference.

5.6. Ablation study

5.6.1. Ablation of surrogate model

We conduct ablation experiments to evaluate the contribution of the Privacy Region Localization component and the sensitivity to surrogate model choice. Results are reported in Table 7 using GPT-4 score on the LLaVA-300K dataset. Full-image protection leads to a notable utility drop. This demonstrates that text-guided localization is essential, and indiscriminate protection degrades visual features that VLMs rely on for accurate inference. We compare three surrogate models for privacy region localization: CLIP, BLIP, and BLIP2. CLIP achieves the highest utility, while BLIP and BLIP2 yield marginally lower scores. The small gap confirms that our method is robust to surrogate model selection; CLIP's slight advantage is attributed to its stronger cross-modal alignment for open vocabulary localization.

5.6.2. Ablation of text noise η

We experiment with models' performance at different values of η in Table 8, which indicates that the model's output scores improve progressively, as the noise level increases. When η exceeds 2, the performance approaches that of the original data.

6. Security analysis and limitations

6.1. Security guarantees

Our method protects against several key threats identified in recent AI security literature. Specifically, when untrained malicious third parties access the protected data, they cannot identify specific sensitive content due to the frequency-domain obfuscation and semantic perturbation applied to privacy-sensitive regions. This privacy-preserving mechanism is particularly relevant in cloud-based VLM services (Deng et al., 2025a,b). The frequency domain transformation using BDCT preserves useful visual features, maintaining model effectiveness on privacy-preserved images.

6.2. Utility–privacy trade-off

Our method balances privacy and utility through two key hyperparameters: entropy threshold ϵ for images and noise parameter η for text. As demonstrated in Table 5, decreasing ϵ enhances privacy protection by removing more frequency-domain information, but significantly degrades utility. At $\epsilon = 0.3$, the GPT-4o score drops to 0.3525 while achieving stronger defense against reconstruction attacks. Conversely, higher ϵ values maintain better utility but weaken privacy guarantees. Similarly, Table 8 and Fig. 6 reveal that text noise parameter η exhibits a comparable trade-off: lower η values are better for utility

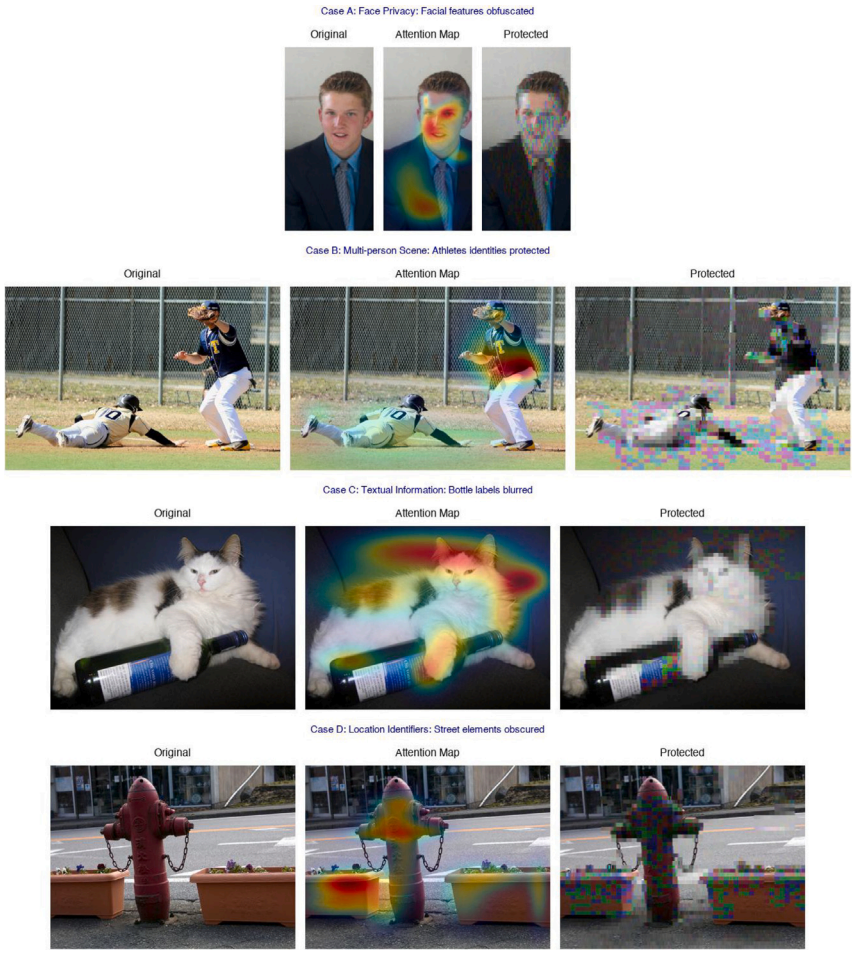


Fig. 10. Qualitative results of privacy-sensitive region identification and protection. Each row shows three stages: the original image (left), the attention map generated by the surrogate VLM under the user's privacy instruction highlighting identified sensitive regions (middle), and the protected image after BDCT-based obfuscation (right).

and higher ASR. The selected configuration $\eta = 2$ achieves near-zero attack success rates while preserving utility. However, users can adjust these parameters based on their specific requirements.

6.3. *Privacy-preserving localization strategy*

POS-based Privacy Token Identification. While noun/pronoun-based POS tagging provides a computationally lightweight and regulatory-aligned approach to privacy token identification (Sec. 4.1), it may miss privacy information embedded in verb phrases or complex contextual descriptions, and is less accurate than directly employing high-performance multimodal models to localize privacy regions in images. However, under computational constraints, where deploying models like DeepSeek (A. Liu et al., 2024) or Qwen3-235B (Yang et al., 2025)on user devices just for localizing privacy regions is impractical, our method represents a near-optimal solution for balancing privacy protection effectiveness and efficiency. In our future work, integrating lightweight models such as Qwen3-VL-2B-Instruct or Gemma3-4B(Team, 2025), potentially combined with RAG (Gao et al., 2023) techniques, would enhance accuracy while preserving computational feasibility for edge deployment.

Limitations of Attention-Based Localization. While our method leverages cross-modal attention maps for privacy-sensitive region localization, two types of localization errors warrant discussion. *False negatives* arise when attention fails to identify genuinely sensitive regions: attention is not explicitly designed for spatial grounding and may miss highly abstract or implicit cues (e.g., background landmarks, contextual identifiers). The visual encoder's fixed 14×14 patch partitioning further limits spatial precision, meaning sub-patch sensitive content may not be perfectly isolated. Regions not flagged by the attention map will not be protected regardless of the entropy threshold. *False positives* occur when attention inadvertently activates over non-sensitive but visually salient regions; our default attention threshold (0.6) mitigates this by restricting protection to high-relevance patches only. The cost of ignoring localization selectivity is directly evidenced by our ablation in Table 7: replacing selective attention-guided protection with uniform full-image protection degrades GPT-4 score from 61.18 to 54.25, confirming that indiscriminate coverage causes substantial utility loss. In practice, the most privacy-critical content (such as personal identifiers, faces, and named entities) tends to produce strong and consistent cross-modal attention responses, which is why our method achieves effective protection in the evaluated scenarios. Future work could explore grounding-aware localization models (e.g., Grounding DINO (S. Liu et al., 2024)) to improve coverage of implicit privacy cues while preserving computational efficiency.

7. Conclusion

This paper presents a privacy protection method for image–text scenarios. We achieve visual and textual data privacy preservation by establishing correspondences between privacy-sensitive regions in images and text. Our method employs a two-stage process: first, using a local surrogate model to localize privacy-sensitive areas in images

based on the user's text. Then we strategically remove frequency channels based on image information entropy. For text, we add Laplace noise in the token vector, and to keep the token's semantics, we add the neighbors' vector to it. Our experiments on MS-COCO and LLaVA-300K demonstrate that models trained on our privacy-preserving datasets maintain at least 96% of the task performance compared to those trained on unprotected data. We validate the robustness of our privacy preservation mechanism by restoring images with restoration models and three attacks for textual data. We plan to extend our method to other modalities, such as audio and text, in future work. We will also commit to enhancing privacy while maintaining practicality.

CRediT authorship contribution statement

Zixiang Liu: Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation. **Chi Chen:** Writing – review & editing, Validation, Supervision, Formal analysis. **Shuguang Yuan:** Writing – review & editing. **Weilong Huang:** Software, Data curation. **Xiaojie Zhu:** Writing – review & editing, Supervision, Formal analysis. **Peizhuo Lv:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Zixiang Liu reports financial support was provided by Chinese Academy of Sciences Institute of Information Engineering. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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